

The Lasso

- The Lasso is a shrinkage method like Ridge with important differences.
- The first justification for its use is again prediction accuracy (bias-variance tradeoff).
- The second is interpretation: with a large number of predictors, we would like to determine a subset of features exhibiting the strongest effect. This is something Ridge was not able to do.
 $\Rightarrow$ *LASSO shrinks some coefficients and sets others to 0, and hence tries to retain the good features of subset selection and ridge.*
- Consider data $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^p$.
- Assume the $\{x_{ij}\}_{i=1, j=1}^{N, p}$ are standardized s.t. $N^{-1} \sum_{i=1}^N x_{ij} = 0$, $N^{-1} \sum_{i=1}^N x_{ij}^2 = 1$. The reason is that, similarly to Ridge, LASSO is not scale invariant, standardization is therefore necessary.
- The LASSO estimate is defined as:

$$(\hat{\alpha}, \hat{\beta})' = \arg \min_{\alpha, \beta} \sum_{i=1}^N \left(y_i - \alpha - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{s.t.} \quad \sum_{j=1}^p |\beta_j| \leq t, \quad (1)$$

where $t \geq 0$ is a tuning parameter.

- $\forall t \geq 0$, $\hat{\alpha}$ is s.t.:

$$\partial_{\alpha} \left(\sum_{i=1}^N \left[y_i - \alpha - \sum_{j=1}^p \beta_j x_{ij} \right]^2 \right) \Big|_{\alpha=\hat{\alpha}} = 0. \quad (2)$$

$$\Rightarrow -2 \sum_{i=1}^N \left(y_i - \hat{\alpha} - \sum_{j=1}^p \beta_j x_{ij} \right) = 0. \quad (3)$$

$$\Rightarrow \sum_{i=1}^N y_i - N\hat{\alpha} - \sum_{i=1}^N \sum_{j=1}^p \beta_j x_{ij} = 0. \quad (4)$$

$$\Leftrightarrow \bar{y} - \hat{\alpha} - \sum_{j=1}^p \left(\beta_j \sum_{i=1}^N x_{ij} N^{-1} \right) = 0. \quad (5)$$

$$\Rightarrow \hat{\alpha} = \bar{y} \quad \forall t \geq 0. \quad (6)$$

Hence we can assume w.l.o.g that $\bar{y} = 0$ and therefore $\hat{\alpha} = 0$, and omit it.

- For the same reasons as Ridge, α is left out of the penalty term.
- Consider the orthonormal case where the design matrix $X \in \mathbb{R}^{N \times p}$ is orthogonal, i.e. $X'X = I_p$.
- In that case, we have: $\hat{\beta}^{\text{ls}} = (X'X)^{-1}X'y = X'y$, which is the argmin of the error sum of squares:

$$\|\underline{y} - X\beta\|^2 = (\underline{y} - X\beta)'(\underline{y} - X\beta) = \underline{y}'\underline{y} - \underline{y}'X\beta - \beta'X'\underline{y} + \beta'X'X\beta \quad (7)$$

$$= \|\underline{y}\|^2 - 2\beta'X'\underline{y} + \beta'\beta \quad (8)$$

$$= \|\underline{y}\|^2 - 2\beta'\hat{\beta}^{\text{ls}} + \beta'\beta \quad (9)$$

$$= \|\underline{y}\|^2 + \hat{\beta}^{\text{ls}'}\hat{\beta}^{\text{ls}} - 2\beta'\hat{\beta}^{\text{ls}} + \beta'\beta - \hat{\beta}^{\text{ls}'}\hat{\beta}^{\text{ls}} \quad (10)$$

$$= \|\underline{y}\|^2 - \|\hat{\beta}^{\text{ls}}\|^2 + \sum_{j=1}^p \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 \quad (11)$$

$$\propto \sum_{j=1}^p \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2, \quad (12)$$

hence the LASSO problem in the orthonormal case is:

$$\min_{\beta} \sum_{j=1}^p \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 \quad \text{s.t.} \quad \sum_{j=1}^p |\beta_j| \leq t. \quad (13)$$

- The associated Lagrangian is:

$$\mathcal{L}(\beta, \lambda) = \sum_{j=1}^p \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 + \lambda \left(\sum_{j=1}^p |\beta_j| - t \right) \quad (14)$$

- Because the problem is convex and the constraint separable, we minimize it coordinate-wise. The subproblems are:

$$\min_{\beta_j} \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 + \lambda |\beta_j|, \quad \forall j = 1, \dots, p. \quad (15)$$

- Take $\beta_j > 0$, then: $\min_{\beta_j} \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 + \lambda \beta_j$

$$\text{F.O.C.: } 2 \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right) + \lambda = 0 \Rightarrow \beta_j = \hat{\beta}_j^{\text{ls}} - \lambda/2, \quad \text{requiring } \hat{\beta}_j^{\text{ls}} > \lambda/2. \quad (16)$$

- Take $\beta_j < 0$, then: $\min_{\beta_j} \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right)^2 - \lambda \beta_j$

$$\text{F.O.C.: } 2 \left(\beta_j - \hat{\beta}_j^{\text{ls}} \right) = \lambda \Rightarrow \beta_j = \hat{\beta}_j^{\text{ls}} + \lambda/2, \quad \text{requiring } \hat{\beta}_j^{\text{ls}} < -\lambda/2. \quad (17)$$

- For $\beta_j = 0$, the subgradient is:

$$0 \in -2\hat{\beta}_j^{\text{ls}} + \lambda \partial|0|. \quad (18)$$

$$\partial|0| \in [-1, 1] \Rightarrow \lambda \partial|0| \in [-\lambda, \lambda] \Rightarrow -2\hat{\beta}_j^{\text{ls}} + \lambda \partial|0| \in [-2\hat{\beta}_j^{\text{ls}} - \lambda, -2\hat{\beta}_j^{\text{ls}} + \lambda].$$

For 0 to be the minimizer, the interval must contain 0, i.e.:

$$-2\hat{\beta}_j^{\text{ls}} - \lambda \leq 0 \leq -2\hat{\beta}_j^{\text{ls}} + \lambda \quad (19)$$

$$\Rightarrow -\lambda/2 \leq \hat{\beta}_j^{\text{ls}} \leq \lambda/2 \iff |\hat{\beta}_j^{\text{ls}}| \leq \lambda/2, \quad \text{hence:} \quad (20)$$

$$\hat{\beta}_j = \begin{cases} \hat{\beta}_j^{\text{ls}} - \lambda/2 & \text{if } \hat{\beta}_j^{\text{ls}} > \lambda/2 \\ 0 & \text{if } |\hat{\beta}_j^{\text{ls}}| \leq \lambda/2, \quad \text{i.e.:} \\ \hat{\beta}_j^{\text{ls}} + \lambda/2 & \text{if } \hat{\beta}_j^{\text{ls}} < -\lambda/2 \end{cases} \quad (21)$$

$$\hat{\beta}_j = \text{sign}(\hat{\beta}_j^{\text{ls}}) \max \left\{ \left| \hat{\beta}_j^{\text{ls}} \right| - \lambda/2, 0 \right\}. \quad (22)$$

For simplicity of notation denote $\lambda/2 = \gamma$ so that:

$$\hat{\beta}_j = \text{sign}(\hat{\beta}_j^{\text{ls}}) \max \left\{ \left| \hat{\beta}_j^{\text{ls}} \right| - \gamma, 0 \right\}, \quad (23)$$

which is the argmin of $\frac{1}{2} (\beta_j - \hat{\beta}_j^{\text{ls}})^2 + \gamma |\beta_j|$ (equivalent).

- To compare the orthonormal setups, Ridge scales the coefficients by a constant factor $((1 + \gamma)^{-1} \hat{\beta}_j^{\text{ls}})$, whereas the LASSO translates by a constant factor, truncating at zero.
- When $\lambda = 0$, the constraint does not bind and intuitively:

$$\hat{\beta}_j = \text{sign}(\hat{\beta}_j^{\text{ls}}) \max \left\{ \left| \hat{\beta}_j^{\text{ls}} \right|, 0 \right\} = \hat{\beta}_j^{\text{ls}}. \quad (24)$$

- The reason why LASSO produces coefficients that are exactly equal to zero is the following. Consider the criterion function:

$$\sum_{i=1}^N (y_i - x_i' \beta)^2 = (\underline{y} - X\beta)'(\underline{y} - X\beta) = \underline{y}'\underline{y} - \underline{y}'X\beta - \beta'X'\underline{y} + \beta'X'X\beta \quad (25)$$

- Consider the following quadratic form:

$$(\beta - \hat{\beta}^{\text{ls}})' X' X (\beta - \hat{\beta}^{\text{ls}}) = \beta' X' X \beta - \beta' X' X \hat{\beta}^{\text{ls}} - \hat{\beta}^{\text{ls}'} X' X \beta \quad (26)$$

$$+ \hat{\beta}^{\text{ls}'} X' X \hat{\beta}^{\text{ls}}. \quad (27)$$

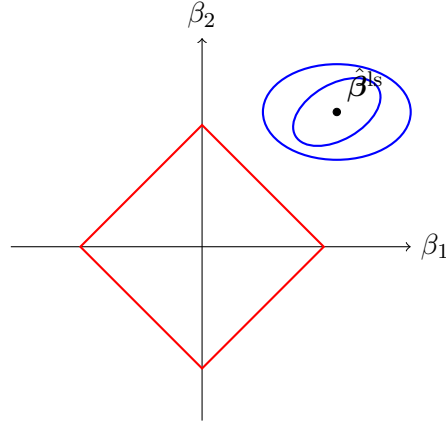
Using the normal equations: $X' X \hat{\beta}^{\text{ls}} = X' \underline{y}$

$$= \beta' X' X \beta - 2\beta' X' \underline{y} + \hat{\beta}^{\text{ls}'} X' X \hat{\beta}^{\text{ls}} \quad (28)$$

$$= \|\underline{y} - X\beta\|^2 - \underline{y}' \underline{y} + \hat{\beta}^{\text{ls}'} X' X \hat{\beta}^{\text{ls}}, \quad (29)$$

hence: $\sum_{i=1}^N (y_i - x'_i \beta)^2 = (\beta - \hat{\beta}^{\text{ls}})' X' X (\beta - \hat{\beta}^{\text{ls}})$, plus some constant terms.

- For $p = 2$, the constraint is: $|\beta_1| + |\beta_2| \leq t$



For ridge, we would get a circle and there would be no corners for the contours to hit and zero solutions will rarely result.

- Keep $p = 2$ and assume $\hat{\beta}_1^{\text{ls}}, \hat{\beta}_2^{\text{ls}} > 0$ so that the constraint is: $\beta_1 + \beta_2 \leq t$ (for the $p = 2$ case, LASSO actually keeps the same sign as LS).
- The RSS is: $(\beta - \hat{\beta}^{\text{ls}})' X'X (\beta - \hat{\beta}^{\text{ls}}) + c$, where c is a constant.
- Because the predictors are standardized:

$$X'X = \begin{bmatrix} x_1 & x_2 \end{bmatrix}' \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{pmatrix} x_1'x_1 & x_1'x_2 \\ x_2'x_1 & x_2'x_2 \end{pmatrix} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \text{where } \rho = \text{Corr}(x_1, x_2), \quad (30)$$

hence:

$$\text{RSS}(\beta) = \begin{pmatrix} \beta_1 - \hat{\beta}_1^{\text{ls}} \\ \beta_2 - \hat{\beta}_2^{\text{ls}} \end{pmatrix}' \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \begin{pmatrix} \beta_1 - \hat{\beta}_1^{\text{ls}} \\ \beta_2 - \hat{\beta}_2^{\text{ls}} \end{pmatrix} \quad (31)$$

$$= \begin{pmatrix} \beta_1 - \hat{\beta}_1^{\text{ls}} & \beta_2 - \hat{\beta}_2^{\text{ls}} \end{pmatrix} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \begin{pmatrix} \beta_1 - \hat{\beta}_1^{\text{ls}} \\ \beta_2 - \hat{\beta}_2^{\text{ls}} \end{pmatrix} \quad (32)$$

$$= \left((\beta_1 - \hat{\beta}_1^{\text{ls}}) + \rho (\beta_2 - \hat{\beta}_2^{\text{ls}}), \rho (\beta_1 - \hat{\beta}_1^{\text{ls}}) + (\beta_2 - \hat{\beta}_2^{\text{ls}}) \right) \begin{pmatrix} \beta_1 - \hat{\beta}_1^{\text{ls}} \\ \beta_2 - \hat{\beta}_2^{\text{ls}} \end{pmatrix} \quad (33)$$

$$\text{RSS}(\beta) = (\beta_1 - \hat{\beta}_1^{\text{ls}})^2 + 2\rho (\beta_1 - \hat{\beta}_1^{\text{ls}}) (\beta_2 - \hat{\beta}_2^{\text{ls}}) + (\beta_2 - \hat{\beta}_2^{\text{ls}})^2 + c, \quad (34)$$

so that the Lagrangean is:

$$\mathcal{L} = (\beta_1 - \hat{\beta}_1^{\text{ls}})^2 + (\beta_2 - \hat{\beta}_2^{\text{ls}})^2 + 2\rho (\beta_1 - \hat{\beta}_1^{\text{ls}}) (\beta_2 - \hat{\beta}_2^{\text{ls}}) + \lambda(\beta_1 + \beta_2 - t). \quad (35)$$

$$\partial_{\beta_1} \mathcal{L} = 0 \iff 2(\beta_1 - \hat{\beta}_1^{\text{ls}}) + 2\rho(\beta_2 - \hat{\beta}_2^{\text{ls}}) + \lambda = 0. \quad (36)$$

$$\iff (\beta_1 - \hat{\beta}_1^{\text{ls}}) + \rho(\beta_2 - \hat{\beta}_2^{\text{ls}}) = -\lambda/2 \quad (*)$$

$$\partial_{\beta_2} \mathcal{L} = 0 \iff \rho(\beta_1 - \hat{\beta}_1^{\text{ls}}) + (\beta_2 - \hat{\beta}_2^{\text{ls}}) = -\lambda/2 \quad (**)$$

- Combining: $(1 - \rho)(\beta_1 - \hat{\beta}_1^{\text{ls}}) - (1 - \rho)(\beta_2 - \hat{\beta}_2^{\text{ls}}) = 0.$

$$\iff (1 - \rho) \left[(\beta_1 - \hat{\beta}_1^{\text{ls}}) - (\beta_2 - \hat{\beta}_2^{\text{ls}}) \right] = 0. \quad (37)$$

- Assuming the absence of multicollinearity, $\rho \neq 1$, so that:

$$\beta_1 - \hat{\beta}_1^{\text{ls}} = \beta_2 - \hat{\beta}_2^{\text{ls}}. \quad (38)$$

- Define γ s.t. $\beta_1 = \hat{\beta}_1^{\text{ls}} - \gamma$ and $\beta_2 = \hat{\beta}_2^{\text{ls}} - \gamma$, so that:

$$\beta_1 - \hat{\beta}_1^{\text{ls}} = -\gamma = \beta_2 - \hat{\beta}_2^{\text{ls}}, \quad \text{so that:} \quad (39)$$

$$\hat{\beta}_j^{\text{LASSO}} = \max \left\{ \hat{\beta}_j^{\text{ls}} - \gamma, 0 \right\} \quad (\text{to keep the same sign}). \quad (40)$$

Hence, when $\rho = 1$, LASSO applies the same amount of shrinkage to both OLS coefficients, regardless of ρ .

Since we have some shrinkage, the constraint will bind: $\beta_1 + \beta_2 = t$. Leading to:

$$\hat{\beta}_1^{\text{ls}} + \hat{\beta}_2^{\text{ls}} - 2\gamma = t \Rightarrow \gamma = \frac{\hat{\beta}_1^{\text{ls}} + \hat{\beta}_2^{\text{ls}} - t}{2}, \quad \text{so that:} \quad (41)$$

$$\hat{\beta}_1^{\text{LASSO}} = \frac{\hat{\beta}_1^{\text{ls}} - \hat{\beta}_2^{\text{ls}} + t}{2} \quad \text{and} \quad \hat{\beta}_2^{\text{LASSO}} = \frac{\hat{\beta}_2^{\text{ls}} - \hat{\beta}_1^{\text{ls}} + t}{2}, \quad \text{i.e.:} \quad (42)$$

$$\hat{\beta}_1^{\text{LASSO}} = \max \left\{ \frac{t}{2} + \frac{\hat{\beta}_1^{\text{ls}} - \hat{\beta}_2^{\text{ls}}}{2}, 0 \right\} \quad (43)$$

$$\hat{\beta}_2^{\text{LASSO}} = \max \left\{ \frac{t}{2} - \frac{\hat{\beta}_1^{\text{ls}} - \hat{\beta}_2^{\text{ls}}}{2}, 0 \right\} \quad (44)$$

In that case, it allocates the L1 budget proportionally to the gap between the LS estimators.

- Due to the non-linearity and non-differentiability of the problem, it is difficult to obtain an accurate estimate of the standard errors.
- One approach is via bootstrap.
- Alternatively, approximate $\sum_{j=1}^p |\beta_j|$ by $\sum_{j=1}^p \beta_j^2 / |\beta_j|$, fixing the denominator at the current LASSO estimate the penalty term becomes a ridge penalty:

$$\lambda \sum_{j=1}^p |\beta_j| \simeq \lambda \sum_{j=1}^p \frac{\beta_j^2}{|\hat{\beta}_j^{\text{LASSO}}|}, \quad \text{for:} \quad (45)$$

$$W = \text{diag} \left(\left\{ |\hat{\beta}_j^{\text{LASSO}}| \right\}_{j=1}^p \right) \quad \text{and denote } W^- \text{ its generalized inverse} \quad (46)$$

(as some coefficients can be set to zero due to the penalty). The zero terms are kept equal to zero in W^- , the other terms are $|\hat{\beta}_j^{\text{LASSO}}|^{-1}$.

- The approximate objective function is then:

$$\|\underline{y} - X\beta\|^2 + \lambda \beta' W^- \beta, \quad \text{the argmin is then similar to Ridge:} \quad (47)$$

$$\hat{\beta}^{\text{LASSO}} \simeq (X'X + \lambda W^-)^{-1} X' \underline{y}, \quad \text{then, assuming} \quad (48)$$

$$\mathbb{E}(\varepsilon \mid X) = 0 \text{ and } \mathbb{V}(\varepsilon \mid X) = \sigma^2 I_N, \text{ we get:} \quad (49)$$

$$\mathbb{V}(\underline{y} \mid X) = \mathbb{V}(X\beta + \varepsilon \mid X) = \mathbb{V}(\varepsilon \mid X) = \sigma^2 I_N, \quad \text{hence:} \quad (50)$$

$$\mathbb{V}(\hat{\beta}^{\text{LASSO}} \mid X) \simeq (X'X + \lambda W^-)^{-1} X' \mathbb{V}(\underline{y} \mid X) X (X'X + \lambda W^-)^{-1} \quad (51)$$

$$= (X'X + \lambda W^-)^{-1} X' X (X'X + \lambda W^-)^{-1} \sigma^2. \quad (52)$$

- Estimated by: $(X'X + \lambda W^-)^{-1} X' X (X'X + \lambda W^-)^{-1} \hat{\sigma}^2$
- The problem being that for $\hat{\beta}_j^{\text{LASSO}} = 0$ we get an estimate of zero.
- Suppose $y = f(x) + \varepsilon$ for $\mathbb{E}(\varepsilon) = 0$ and $\mathbb{V}(\varepsilon) = \sigma^2$. The MSE of an estimate $\hat{\eta}(x)$ is:

$$\text{MSE} = \mathbb{E}[(\hat{\eta}(x) - \eta(x))^2] \quad (53)$$

- The prediction error is:

$$\text{PE} = \mathbb{E}[(y - \hat{\eta}(x))^2] = \mathbb{E}[(\eta(x) + \varepsilon - \hat{\eta}(x))^2] \quad (54)$$

$$= \mathbb{E}[(\hat{\eta}(x) - \eta(x))^2] + \mathbb{E}(\varepsilon^2) + 2\mathbb{E}[\varepsilon (\hat{\eta}(x) - \eta(x))] \quad (55)$$

$$\text{PE} = \text{MSE} + \sigma^2, \quad \text{assuming } \varepsilon \text{ mean independent of } x. \quad (56)$$

- Consider the model $\eta(x) = x'\beta$, then:

$$\text{MSE} = \mathbb{E}[(X\hat{\beta} - X\beta)'(X\hat{\beta} - X\beta)] \quad (57)$$

$$= (\hat{\beta} - \beta)' \mathbb{E}(X'X) (\hat{\beta} - \beta). \quad (58)$$

- Consider $s = t / \sum_{j=1}^p |\hat{\beta}_j^{\text{ls}}|$, when $s = 1$, $\sum_{j=1}^p |\hat{\beta}_j^{\text{ls}}| = t$ and there is no shrinkage. When $s = 0$, $t = 0$ so that all coefficients are zero.

- We therefore perform a search over $s \in [0, 1]$.
- By CV, say with $K = 5$. Let $\kappa : \{1, \dots, N\} \rightarrow \{1, \dots, 5\}$ the partition indicating to which fold is allocated observation i .
- We then look at the CV estimate of prediction error:

$$\text{CV}(\hat{\eta}, s) = \frac{1}{N} \sum_{i=1}^N \left(y_i - \hat{\eta}^{-\kappa(i)}(x_i) \right)^2 \quad (59)$$

where $\hat{\eta}^{-\kappa}(x)$ is the fitted function computed with the κ -th part of the data removed.

- We then select $\hat{s} = \arg \min_s \text{CV}(\hat{\eta}, s)$ and recover $\hat{t} = \hat{s} \sum_{j=1}^p \left| \hat{\beta}_j^{\text{ls}} \right|$.
- This is computationally intensive; consider again the formula for Generalized Cross-Validation derived in Section 7:

$$\text{GCV}(\hat{\eta}) = \frac{1}{N} \sum_{i=1}^N \frac{(y_i - \hat{\eta}(x_i))^2}{(1 - \text{Tr}(S)/N)^2}, \quad \text{where } S \text{ is a matrix s.t. } \hat{\eta} \equiv \underline{\hat{y}} = S\underline{y}, \quad (60)$$

which is an approximation of LOOCV that can be calculated with just a single model fit.

- Then consider again the approximation to the penalty $\sum_{j=1}^p \beta_j^2 / |\hat{\beta}_j^{\text{LASSO}}|$, so that:

$$\text{RSS}(\beta) = \|\underline{y} - X\beta\|^2 + \lambda\beta'W^-\beta \quad \text{and therefore:} \quad (61)$$

$$\tilde{\beta} = (X'X + \lambda W^-)^{-1} X' \underline{y} \equiv S\underline{y}, \quad \text{so that the effective degrees of freedom are:} \quad (62)$$

$$\text{Tr}(S) = \text{Tr}[(X'X + \lambda W^-)^{-1} X'] \equiv \rho(t) \quad \text{as } \lambda \text{ depends on } t. \quad (63)$$

- And then $\text{GCV}(t) = \frac{1}{N} \sum_{i=1}^N \frac{(y_i - \hat{\eta}(x_i))^2}{(1 - \rho(t)/N)^2}$

$$= \frac{1}{N} \frac{\text{RSS}(t)}{(1 - \rho(t)/N)^2} \quad (64)$$

- The LASSO can also be seen as the Bayes posterior mode under independent double-exponential priors.
- Consider the following prior distribution for the coefficients:

$$p(\beta_j; \tau) = \frac{1}{2\tau} \exp(-|\beta_j|/\tau), \quad \text{where } \tau \text{ is a scale parameter.} \quad (65)$$

Assuming they are i.i.d.:

$$p(\beta; \tau) = \prod_{j=1}^p p(\beta_j; \tau) = \left(\frac{1}{2\tau}\right)^p \exp\left(-\frac{1}{\tau} \sum_{j=1}^p |\beta_j|\right) \quad (66)$$

- Assuming a linear model $y_i = x_i' \beta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the likelihood for the observed data is proportional to:

$$p(\underline{y} \mid \beta, \sigma) \propto \exp\left(-\frac{1}{2} \|\underline{y} - X\beta\|^2\right) \quad (67)$$

- The posterior is then:

$$p(\beta \mid \underline{y}; \sigma^2, \tau) = p(\underline{y} \mid \beta, \sigma^2) p(\beta; \tau) / p(\underline{y}) \quad (68)$$

$$\propto p(\underline{y} \mid \beta, \sigma^2) p(\beta; \tau) \quad (69)$$

$$\Rightarrow \log(p(\beta \mid \underline{y}; \sigma^2, \tau)) \propto \log p(\underline{y} \mid \beta, \sigma^2) + \log p(\beta; \tau) \propto -\frac{1}{2} \|\underline{y} - X\beta\|^2 - \frac{1}{\tau} \sum_{j=1}^p |\beta_j|. \quad (70)$$

The MAP estimator is then:

$$\hat{\beta}^{\text{MAP}} = \arg \max_{\beta} \{p(\beta \mid \underline{y}; \sigma^2, \tau)\} = \arg \min_{\beta} \left\{ \frac{1}{2} \|\underline{y} - X\beta\|^2 + \frac{1}{\tau} \sum_{j=1}^p |\beta_j| \right\}, \quad \text{hence the equivalence with } 1/\tau = \lambda. \quad (71)$$

- Now, we need some algorithms to find solutions to the LASSO problem.
- Fix $t \geq 0$, then the problem

$$\min_{\beta} \sum_{i=1}^N (y_i - x'_i \beta)^2 \quad \text{s.t.} \quad \sum_{j=1}^p |\beta_j| \leq t \quad (72)$$

is a LS problem with 2^p inequality constraints as each β_j can be positive or negative.

- Let $g(\beta) = \sum_{i=1}^N (y_i - x'_i \beta)^2$. Let δ_i be a p -tuple representing one sign pattern $i = 1, 2, \dots, 2^p$, $\delta_i = (\pm 1, \pm 1, \dots, \pm 1)$. If β follows the sign pattern of δ_i then $\sum_{j=1}^p |\beta_j| = \delta'_i \beta$, so that the constraint becomes $\delta'_i \beta \leq t$.
- But we don't know the true sign pattern in advance, so all of these inequalities must hold simultaneously, i.e.:

$$\delta'_1 \beta \leq t, \quad \delta'_2 \beta \leq t, \quad \dots, \quad \delta'_{2^p} \beta \leq t \quad (73)$$

- For given β , let $E = \{i : \delta'_i \beta = t\}$ the equality set (binding) and $S = \{i : \delta'_i \beta < t\}$ the slack set.
- Let G_E be the matrix whose rows are δ'_i , $\forall i \in E$.
- Let $\frac{1}{\#G_E}$ a vector of ones of length equal to the number of rows of G_E .

- Start with $\delta_{i_0} = \text{sgn}(\hat{\beta}^{\text{ls}})$ and $E = \{i_0\}$.
- Solve the constrained LS problem w/ constraint $\delta'_{i_0} \beta \leq t$. If $\sum_{j=1}^p \left| \hat{\beta}_j^{\text{CLS},0} \right| \leq t$, then we are done. Otherwise, take i_1 s.t. $\delta_{i_1} = \text{sgn}(\hat{\beta}^{\text{CLS},0})$ where $\hat{\beta}^{\text{CLS}}$ and solve the CLS problem with $\delta'_{i_1} \beta \leq t$, if $\sum_{j=1}^p \left| \hat{\beta}_j^{\text{CLS},1} \right| \leq t$, then we are done, otherwise repeat...

Inference after model selection.

- Many authors have shown the ability of the LASSO to recover the correct model as N and p grow. For instance Donoho (2006) focuses on the $p > N$ case and considers the LASSO solution as the bound on t grows.
- He shows that under some assumptions on X , if the true model is sparse, the solution identifies the correct predictors with high probability.
- Let S the set of predictors truly belonging to the model (whose coefficients are non-null) and S^c the set of predictors whose coefficients are zero. The LASSO's job would be to distinguish variables belonging to S from those belonging to S^c .
- Consider X_S and X_{S^c} the design matrices for these two types of variables. For some $j \in S^c$ (a variable that should not be selected), imagine fitting:

$$x_j = X_S \hat{\gamma}_j, \quad \text{the LS estimate is } \hat{\gamma}_j = (X_S' X_S)^{-1} X_S' x_j \quad (74)$$

$$\Rightarrow \hat{\gamma}'_j = x_j' X_S (X_S' X_S)^{-1} \quad (75)$$

- Many of the results in this area assume a condition of the type:

$$\max_{j \in S^c} \|\hat{\gamma}'_j\|_1 \leq (1 - \varepsilon) \quad \text{for some } \varepsilon \in (0, 1], \quad (76)$$

where $\|\cdot\|_1$ is the L1 norm. Meaning that every irrelevant variable must be sufficiently different from the relevant variables, otherwise the LASSO would have trouble distinguishing them.

- When estimating a structural effect, researchers typically face the problem of choosing which controls to include and what is the functional form.
- Consider the model:

$$\begin{cases} y_i = \alpha_0 d_i + g(z_i) + \xi_i, & \mathbb{E}(\xi_i \mid d_i, z_i) = 0 \\ d_i = m(z_i) + v_i, & \mathbb{E}(v_i \mid z_i) = 0 \end{cases} \quad (*)$$

We are interested in the impact α_0 of the treatment d_i . z_i are controls affecting both the policy variable through $m(z_i)$ and the outcome through $g(z_i)$; these functions are unknown.

- We use a linear index to approximate these functions:

$$g(z_i) \simeq x_i' \beta_{g0} + r_{gi}, \quad m(z_i) \simeq x_i' \beta_{m0} + r_{mi}, \quad (77)$$

where r_{gi} and r_{mi} are the approximation errors.

- Substituting in (*):

$$\begin{cases} y_i = \alpha_0 d_i + x_i' \beta_{g0} + r_{gi} + \xi_i \\ d_i = x_i' \beta_{m0} + r_{mi} + v_i. \end{cases} \quad (78)$$

- We consider $x_i \in \mathbb{R}^{p_m}$ where $p = p_m$ can be large relative to sample size N (high-dimensional setup).
- Out of these p controls assume only s of them are sufficient to approximate $g(\cdot)$ and $m(\cdot)$ “correctly”.

- Write the reduced form of y_i :

$$y_i = \alpha_0 (m(z_i) + v_i) + g(z_i) + \xi_i \quad (79)$$

$$= \alpha_0 (x'_i \beta_{m0} + r_{mi} + v_i) + x'_i \beta_{g0} + r_{gi} + \xi_i \quad (80)$$

$$= x'_i (\alpha_0 \beta_{m0} + \beta_{g0}) + (\alpha_0 r_{mi} + r_{gi}) + (\alpha_0 v_i + \xi_i) \quad (81)$$

$$\Rightarrow \begin{cases} y_i = x'_i \bar{\beta}_0 + \bar{r}_i + \bar{\xi}_i & (i) \\ d_i = x'_i \beta_{m0} + r_{mi} + v_i & (ii) \end{cases}, \quad \text{where} \quad \begin{aligned} \bar{\beta}_0 &= \alpha_0 \beta_{m0} + \beta_{g0} \\ \bar{r}_i &= \alpha_0 r_{mi} + r_{gi} \\ \bar{\xi}_i &= \alpha_0 v_i + \xi_i \end{aligned} \quad (82)$$

- The key method is the following (post-double-selection):

(a) Apply LASSO to (ii) and (i) and denote $\hat{I}_1 = \text{Supp}(\hat{\beta}_m^{\text{LASSO}})$ and $\hat{I}_2 = \text{Supp}(\hat{\beta}^{\text{LASSO}})$ (assuming no approximation error). Let W_i be the vector of variables belonging to $\hat{I}_1 \cup \hat{I}_2$.

(b) Estimate $y_i = \alpha_0 d_i + W'_i \gamma + \varepsilon_i$.

Then $\sqrt{N}(\hat{\alpha}_0 - \alpha_0) \xrightarrow{d} \mathcal{N}(0, V)$ for some V . (Belloni, Chernozhukov, Hansen – 2014).

- The intuition behind this double selection approach is the following. Consider the approach of solving for:

$$(\check{\alpha}_0, \check{\beta}_g) = \arg \min_{\alpha_0, \beta_g} \sum_{i=1}^N (y_i - \alpha_0 d_i - x'_i \beta_g)^2 + \lambda \sum_{j=1}^p |\beta_{gj}| \quad (**)$$

In the $p = 2$ case, we have shown that the non-zero estimates are shrunk toward zero and therefore biased, and generally inconsistent.

- Another approach would be OLS post-LASSO by keeping only the non-zero solutions of (**) (without d_i on the RHS).
- The problem is that x_j might be such that $\beta_{jg} \notin \text{Supp}(\hat{\beta}_g)$ but yet $\text{Cov}(x_{ij}, d_i) \neq 0$ and $\beta_{jg0} \neq 0$, leading to omitted variable bias.
- Hence, the double selection approach allows us to keep regressors that impact d_i and/or y_i .